

MACROSCOPIC AHARONOV–BOHM EFFECT AT *L*-BAND MICROWAVE FREQUENCIES

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Experimental tests using a TM_{01} mode circular waveguide verify the Aharonov–Bohm effect at *L*-band frequencies. Additional tests on a free-space range verify longitudinal vector potential wave propagation across 1.5 meters out to a range of 25 meters and beyond. We verify the Lorenz gauge as the only physically correct gauge for these propagating vector potential $\mathbf{A}$ waves.

Keywords: Electromagnetic potential; classical electrodynamics; low-temperature plasma; Aharonov–Bohm effect.

1. Introduction

The Aharonov–Bohm effect¹ has been experimentally verified many times with a static $\mathbf{A}$ potential and dual electron beams,^{2–9} forming an electron-wave interference pattern which shifts from the central peak when $\mathbf{A}$ is applied between the parallel electron beams. The intent of this experiment is to demonstrate momentum transfer due to a microwave $\mathbf{A}$ potential interacting with plasma electrons. To our knowledge, this has never been done before.

2. Experimental Arrangement and Theory

Figure 1 shows the experimental arrangement schematically. A circular copper waveguide 3 meters long (diameter = 17.78 cm = 7.0 inches) is excited in TM_{01} mode at one-end with a centered coaxial longitudinal probe at frequencies between 1280 MHz and 1380 MHz. At the far-end of the guide, a folded (mercury vapor) plasma tube ionized with 200 mA DC current is used as a detector.

The intent here is to send longitudinal E_z and A_z waves down the length of the waveguide. The transmitting longitudinal monopole (14 cm long) is a quarter guide wavelength at 1400 MHz, the upper frequency limit. The lower frequency limit is formed by the waveguide TM_{01} cut-off frequency of 1291 MHz. The TM_{01} mode diagram in circular waveguide is shown in Fig. 2 from Marcuvitz.¹⁰

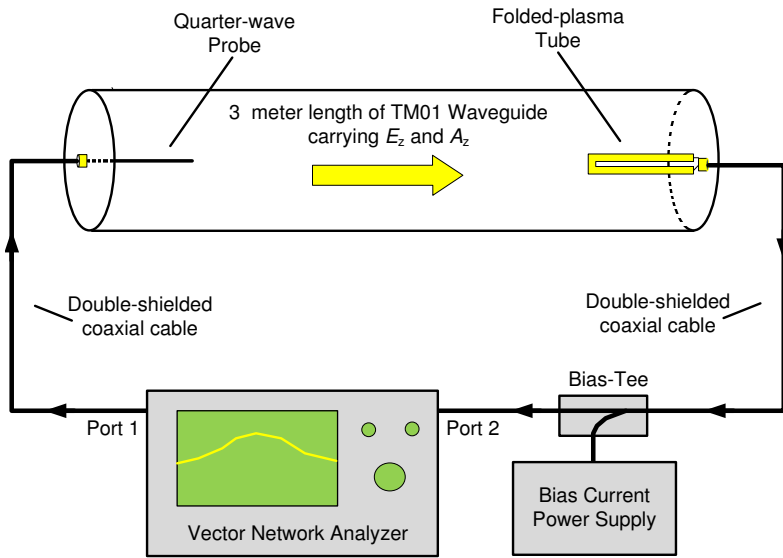


Fig. 1. Experimental arrangement for transmitting and receiving E_z and A_z waves in TM_{01} waveguide.

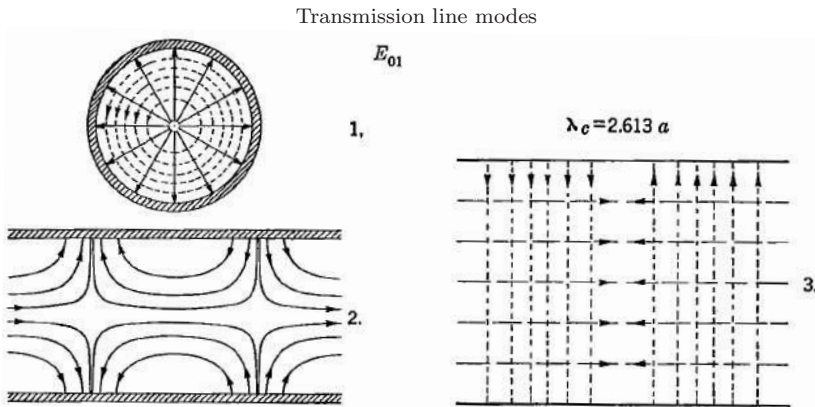


Fig. 2. Modal diagram for TM_{01} waves in circular waveguide, from Marcuvitz.¹⁰

The folded plasma tube detector is shown in Fig. 3. A DC current-regulated power supply is used to ionize the tube and maintain the tube current at 200 mA. The power supply is floating so that a DPDT switch in the bias line can be used to reverse the bias current. This current is fed to the plasma tube by way of a bias-tee^a inserted in the coaxial line leading to the receive port (port 2) of a vector network analyzer (VNA). We define the tube current as *negative* when the coaxial

^aThe bias-tee was made by Microwave Engineering Corporation, MA, USA. The return loss is < -25 dB from 1 GHz to 1.48 GHz. The RF decoupling to the bias input is < -25 dB and the RF insertion loss is -0.04 dB.



Fig. 3. Compact fluorescent lamp serving as a folded plasma tube antenna/detector. The tube is rated at 10 Watts and is 11 cm in length.

center conductor carries negative voltage with respect to ground (coaxial shield). The transmitting port (port 1) of the VNA is connected directly to the coaxial longitudinal probe at the head of the waveguide.

The detector tube operates simultaneously in two modes of operation. First, it operates as a standard folded-monopole plasma antenna to receive E_z in the waveguide. Second, it operates as an Aharonov–Bohm differential detector for A_z in the waveguide. The operation of a “static” Aharonov–Bohm detector is detailed in Fig. 4(a). This is the experimental apparatus that has been widely reported. The plasma tube “dynamic” detector is shown in Fig. 4(b). The incoming vector potential (from the left) couples to the high velocity electron flux in the tube, contributing a differential linear momentum Δp to the electrons. This differential momentum may be coupled to an external $50\ \Omega$ circuit to develop an RF current at frequency F , which is the frequency of the applied vector potential $\mathbf{A}$.

The energy to vary the electron velocity and momentum, the “interaction energy”,¹¹ is equal to

$$U = e(\phi - \mathbf{v} \cdot \mathbf{A}), \quad (1)$$

where e is the electron charge, ϕ is the electric scalar potential, $\mathbf{v}$ is the electron drift velocity in the plasma, and $\mathbf{A}$ is the vector potential in Wb/m (SI units). The vector potential we are using is essentially curl-free (irrotational), $\hat{z}$ -directed, and carries no energy. (The electron drift velocity¹² in the plasma tube is estimated at 40 km/s.) Looking carefully at Eq. (1), we see that the term $\mathbf{v} \cdot \mathbf{A}$ is opposite in the two legs of the plasma tube, as the drift velocity is opposite in these two legs. At any one time, if this portion of the interaction energy is positive in one tube leg, it is negative in the opposite leg. It is this portion of the interaction energy

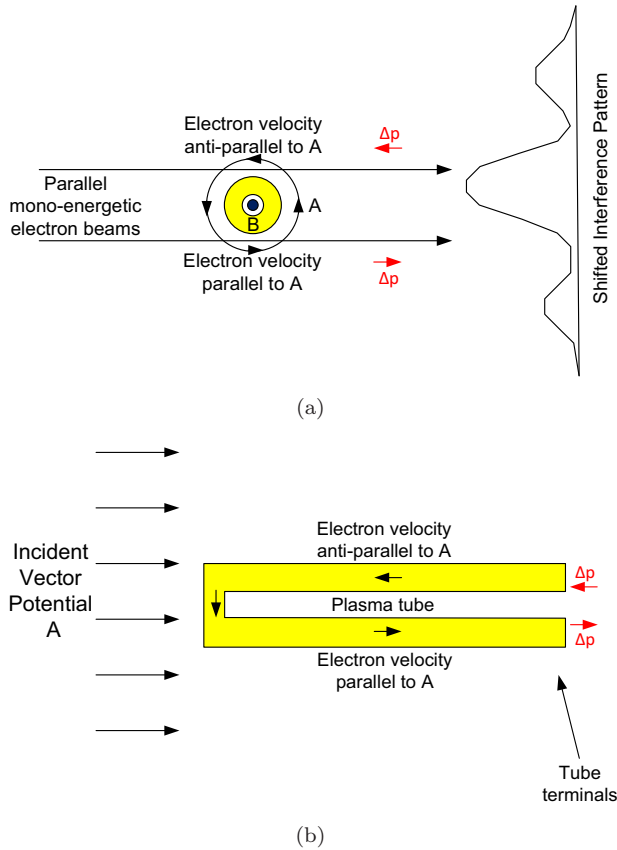


Fig. 4. (a) End-on view of solenoid carrying static magnetic field $\mathbf{B}$. Around the solenoid are concentric static rings of magnetic vector potential $\mathbf{A}$. Two parallel mono-energetic electron beams pass near the solenoid. The electron beam velocity above the solenoid is anti-parallel with $\mathbf{A}$, and has its momentum differentially decreased by Δp to the left. The electron beam velocity below the solenoid is parallel with $\mathbf{A}$, and has its momentum differentially increased by Δp to the right. The two beams, with a momentum difference of $2\Delta p$, interfere on a screen at the right yielding a shifted interference pattern. (b) U-shaped plasma tube carrying high velocity electron flux in the direction indicated by the arrows within the tube. From the left approaches a dynamic magnetic vector potential $\mathbf{A}$, oscillating in time at frequency F (shown at an instant in time). The electron flux in the upper leg of the tube is anti-parallel with potential A_z , and suffers a differential change in momentum to the left of Δp . The electron flux in the lower leg of the tube is parallel with potential A_z , and suffers a differential change in momentum to the right of Δp . At the tube terminals, the electron flux (now electron current) at frequency F may be allowed to pass through an external $50\ \Omega$ circuit, analogous to the interference screen in Fig. 4(a). (The momentum increment Δp occurs at frequency F .)

which can yield positive and negative Δp in the plasma tube legs and terminals. The theoretical basis for momentum transfer to the plasma electrons is presented in Appendix A.

The RF modulation of the DC momentum and velocity forms a traveling TEM wave. (The characteristic impedance of the plasma tube, as a parallel transmission

line, is $125\ \Omega$.) The RF current in the terminals of the tube will be proportional to A_z magnitude and phase. The RF signal travels the length of the plasma tube, experiences a partial voltage reflection at the coaxial interface of $\Gamma = (Z_1 - Z_2)/(Z_1 + Z_2) = (50 - 125)/(50 + 125) = -0.43$, passes through the bias-tee, and into the receive port of the VNA. Reversing the DC bias current polarity with the DPDT switch reverses the current in the respective legs of the tube, and results in a 180° phase shift of the A_z detected signal, without affecting the received signal due to the E_z RF field in the waveguide (receiving as a folded-monopole antenna).

3. Experimental Results for Waveguide Propagation

The experimental data are taken with a full 2-port calibration of the VNA and cables with reference loads, shorts, and opens traceable to NIST. The experimental data, across the band 1280 MHz–1380 MHz, are shown in Fig. 5(a). At the top of the plot is $|S_{11}|^2$, below this is $(1 - |S_{11}|^2)$, and at the bottom is $|S_{21}|^2$ for both positive and negative tube bias current. When losses are present, $(1 - |S_{11}|^2)$ exceeds $|S_{21}|^2$. For S21 negative bias, the A_z and E_z superpose linearly in the tube to yield a resultant signal with a mean value of $-16.5\ \text{dB} \pm 1.0\ \text{dB}$. For S21 positive bias, A_z and E_z superpose out-of-phase to yield a resultant signal some 5 dB lower. Note that both the $|S_{21}|^2$ curves are *anti-correlated* with the $(1 - |S_{11}|^2)$ curve. Only the contribution of A_z can make this possible. When the waveguide is an integral number of half wavelengths, the short circuit (flange) at the end of the guide will be transferred to the monopole at the head of the guide. The monopole will have a reflection coefficient of $\Gamma = -1$ back into the coaxial feedline from the VNA. The monopole current will be twice the current for a matched load, producing the maximum vector potential at that particular wavelength, which is detected by the plasma tube as a maximum response ... hence the anti-correlation between $(1 - |S_{11}|^2)$ and $|S_{21}|^2$, both for negative and positive bias current.

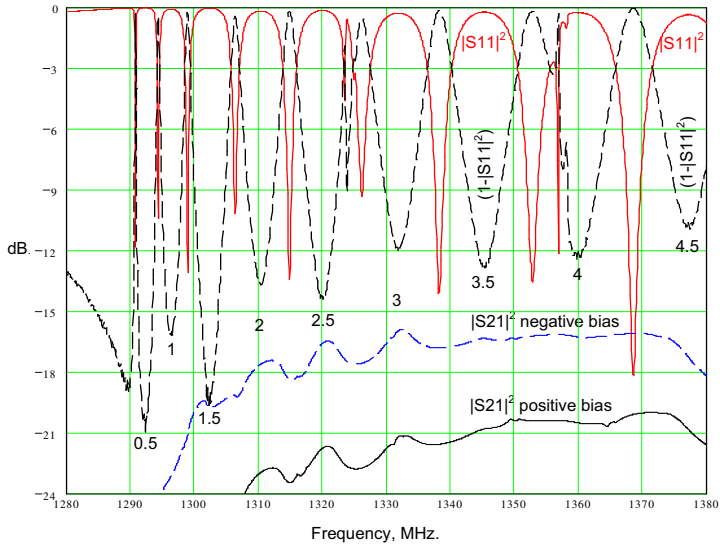
The mean difference between the two $|S_{21}|^2$ curves in Fig. 5(a) is $\Delta = 5\ \text{dB}$, and can be used to determine the relative response of the plasma tube to E_z and A_z . We have

$$\Delta = 10 \log \left[\frac{E_z + A_z}{E_z - A_z} \right]^2 = 5\ \text{dB}. \quad (2)$$

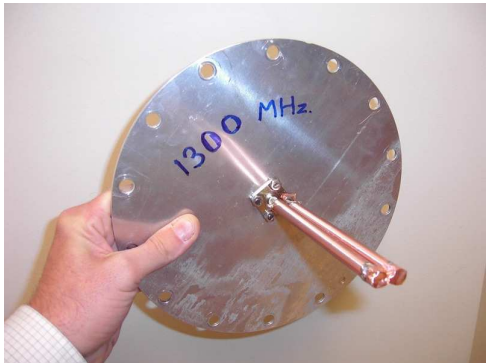
Therefore, $|A_z/E_z| = 0.3$. In the following section, we verify that E_z is indeed the larger response.

A copper mockup of the plasma tube was built and tested in the waveguide as in Fig. 5(b), rather than the plasma tube. $|S_{21}|^2$ does not bifurcate with current, and is well-correlated with $(1 - |S_{11}|^2)$. The metallic antenna results are shown in Fig. 5(c).

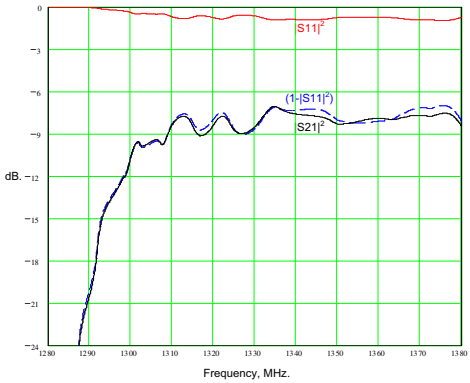
As might be expected, there is a minimal Interaction Energy requirement for full momentum transfer from the vector potential to the plasma electrons. This means there is a requirement for at least a minimum electron drift velocity in the plasma for momentum transfer. This is discussed fully in Appendix B.



(a)



(b)



(c)

Fig. 5. (a) Measured $|S_{11}|^2$, calculated $(1 - |S_{11}|^2)$, and measured $|S_{21}|^2$ for a 3-meter length of TM_{01} circular waveguide with a plasma antenna/detector. The TM_{01} cutoff frequency is 1291 MHz. The numbers in the plot refer to the guide length measured in guide wavelengths. (b) Copper mockup of plasma tube geometry. (c) Measured $|S_{11}|^2$, calculated $(1 - |S_{11}|^2)$, and measured $|S_{21}|^2$ for a 3-meter length of TM_{01} circular waveguide with a folded copper monopole as a receiving antenna. $|S_{21}|^2$ is well-correlated with $(1 - |S_{11}|^2)$. The TM_{01} cut-off frequency is 1291 MHz.

4. Free-Space Tests

The tests described above lead directly to the possibility of measuring A_z and E_z signals individually by taking phase data along with the signal amplitudes. In point, if we denote complex S_{21} negative bias as S_{21}^- , and complex S_{21} positive bias as

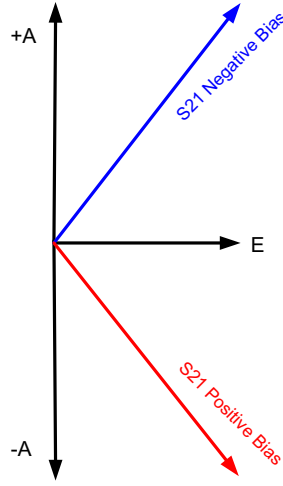


Fig. 6. Vector relationship in time domain between positive bias S21 and negative bias S21.

S21⁺, we may calculate the sum and difference signals:

$$E_z = (S21^- + S21^+)/2, \quad (3a)$$

$$A_z = (S21^- - S21^+)/2. \quad (3b)$$

Of course, S21 is dimensionless, while $\mathbf{E}$ and $\mathbf{A}$ have different dimensions (volts/m and webers/m, respectively). This does not pose a problem, for actually we are referring to the *response* of the plasma tube to $\mathbf{E}$ and $\mathbf{A}$ (rather than $\mathbf{E}$ and $\mathbf{A}$ themselves). In free-space, $\mathbf{E}$ and $\mathbf{A}$ are in time-quadrature according to

$$\mathbf{E} = -\nabla\phi - j\omega\mathbf{A} \quad (4)$$

where ϕ is the electric scalar potential. At 1 m range from the transmitting waveguide aperture (1400 MHz), the $\nabla\phi$ term amounts to 4% of the $j\omega\mathbf{A}$ term and even less at further range; accordingly, $\mathbf{E}$ and $\mathbf{A}$ are in excellent phase quadrature. This was observed in all three free-space tests. In the time domain, the relationship is as in Fig. 6. The relative angle between the positive bias and negative bias terms is shown as $2 \operatorname{atan}(|A/E|)$. Calculated and observed angles are shown in Fig. 7.

Resolving the respective components in this manner allows one to examine how the components depend on distance of propagation. Tests were made with open-ended waveguides at 1.5, 6, and 25 meters, while taking phase data. The waveguides are shown in Figs. 8(a) and 8(b). Consider the case at 1.5 m range. The coupled electron momentum due to A_z is¹³

$$p_z = eA_z = e \left[\frac{\mu}{4\pi} \int_0^L \frac{I(z)e^{jk \cdot r} dz}{|r|} \right], \quad (5)$$

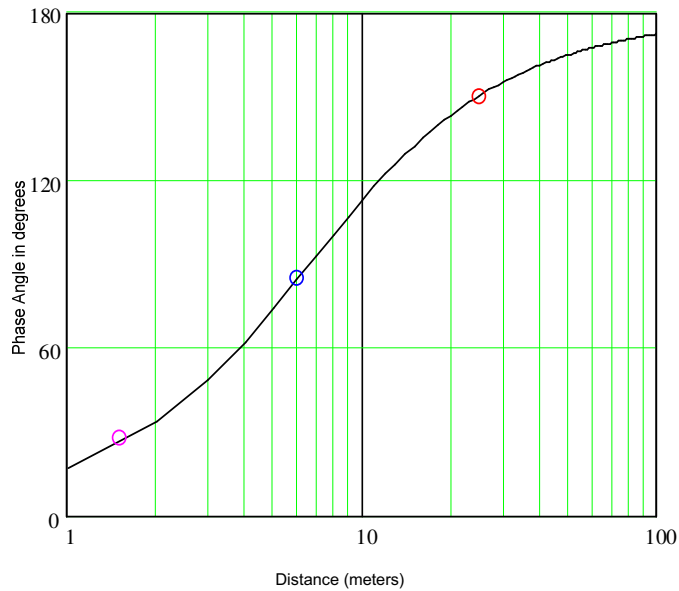


Fig. 7. Calculated and measured angle between positive bias S21 and negative bias S21. For this diagram, the following values for A_z and E_z responses are used: $A_z = 0.00025/R$, $E_z = 0.0016/R^2$ where R is in meters.

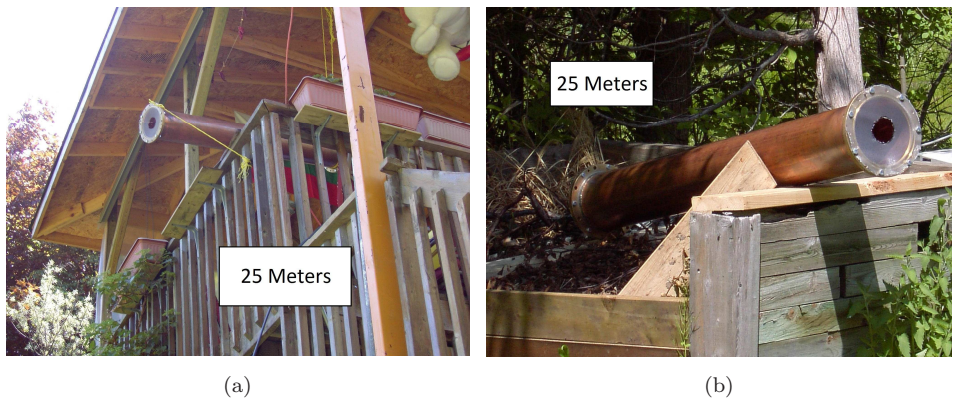


Fig. 8. (a) Transmitting waveguide, TM_{01} mode, for 25 meter propagation test. An aluminum screen on the open-end of the waveguide (with a 7 cm diameter aperture) spreads the on-axis TEM null to approximately 30° full-beamwidth. There is no transmitted TEM signal on-axis. (b) Receiving waveguide, TM_{01} mode, for 25 meter propagation test. An aluminum screen on the open-end of the waveguide (with a 7 cm diameter aperture) spreads the on-axis TEM null to approximately 30° full-beamwidth. There is no reception of TEM signals on-axis.

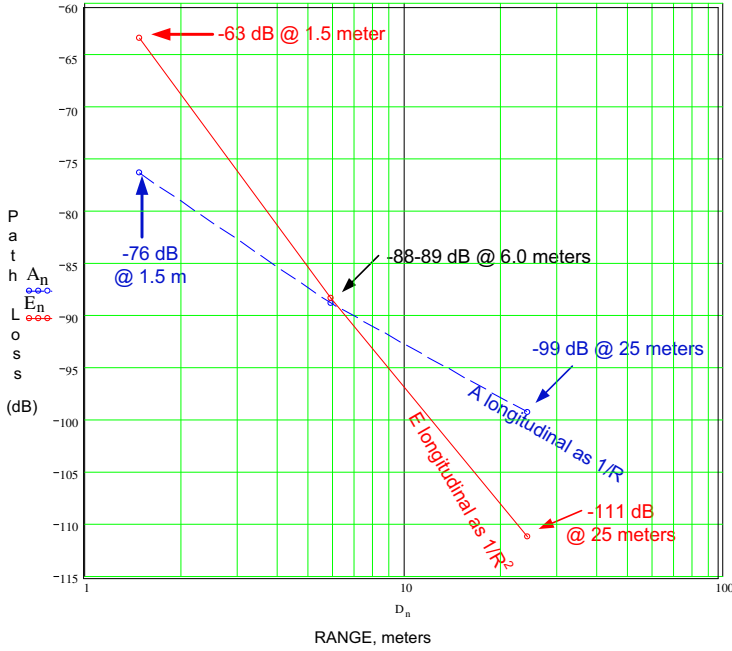


Fig. 9. By using the algorithm in Eqs. (3a) and (3b), the received signal for various propagation distances can be broken into A_z and E_z component values for $|S_{21}|^2$. The two traces are parametrized in Fig. 7 caption.

where e is the electronic charge (1.602×10^{-19} C), I is the retarded current distribution on the monopole due to 10 mW applied power into an impedance $Z \ll 50 \Omega$ (28 mA maximum for a short connected to a 50Ω source), and the integral expression is the usual defining integral¹⁴ for $\mathbf{A}$. We may equate this momentum to $m_e v_e$, where v_e is the RF portion of the electron velocity

$$p_z = eA_z = m_e v_e, \quad (6)$$

and m_e is the electron rest mass (9.11×10^{-31} kg). For the given A_z of about 10^{-10} Wb/m, the RF component of the electron velocity is

$$v_e = (e/m_e)A_z = \pm 20 \text{ m/s}. \quad (7)$$

This magnitude velocity is far higher than can be supported in copper antennas, where the maximum free-electron velocity (due to crystalline lattice scattering) is about 3 mm/s. Accordingly, a plasma detector *must* be used, as no detected signal would be anticipated from a copper antenna.

Experimental results are shown in Fig. 9. E_z propagates as $1/R^2$ which may be anticipated by examining the radiation equation for E_z from a short dipole,¹⁵

$$E_r = \frac{I_0 L \cos \theta e^{j\omega(t-\frac{r}{c})}}{2\pi\epsilon} \left(\frac{1}{cr^2} + \frac{1}{j\omega r^3} \right). \quad (8)$$

A_z propagates as $1/R$ in accordance with the defining integral in Eq. (5). If charge/current continuity is assumed in the transmitting monopole radiator, Eq. (5) immediately determines the proper vector potential gauge as the Lorenz gauge,

$$\nabla \cdot \mathbf{A} = -\mu_0 \varepsilon_0 \partial \phi / \partial t = -(1/c^2) \partial \phi / \partial t, \quad (9)$$

where ϕ is the electric scalar potential. This supports wave propagation at velocity c .

Note that in the laboratory waveguide test, a VNA power of +10 dBm (10 mW) is used. This power level is also used in the 1.5 meter propagation test. An external power amplifier delivering +20 dBm (100 mW) is used for tests at 6 and 25 meters.

5. Detector Responsivity

The plasma detector provides an RF current directly proportional to the vector potential magnitude. The detector *Responsivity* is easy to calculate from the plasma electron concentration and the tube dimensions. The expression starts with the A_z modulated electron velocity in expression (7). This oscillating velocity applies to all free-electrons in the plasma. The electron concentration in the cold plasma is given by the plasma model in Ref. 12, and is about $[n] = 2 \times 10^{17}/\text{m}^3$. The tube radius is 0.007 m. The tube electron current due to A_z is then

$$I_{rf} = \left(\frac{e}{m_e} \right) A_z [n] e \pi r^2. \quad (10)$$

This is $I_{rf}(\text{peak})$ as a function of $A_z(\text{peak})$. To obtain rms current, multiply by $1/\sqrt{2} = 0.707$. There is an additional factor: A_z drops off within the plasma from the external value due to skin effect, documented in Ref. 12, yielding an additional factor of 0.67. With these factors, the detector *Responsivity* = $I_{rf}(\text{rms})/A_z(\text{peak}) = 420 \times 10^3 \text{ A/Wb/m}$. (See the associated discussion in Appendix B.) The available power is $[\text{Resp.} \cdot A_z(\text{peak})]^2 Z_0$, where Z_0 is the impedance of the plasma tube as a parallel transmission line, 125 Ω . To obtain the delivered power to another impedance, multiply by $(1 - \Gamma^2)$.

6. Conclusion

These experiments show a wealth of information that was made possible because we used a simple plasma tube as a detector.

- (1) A folded plasma tube can receive E_z as a simple folded monopole antenna.
- (2) A folded plasma tube can detect A_z as a differential Aharonov–Bohm detector.
- (3) The Aharonov–Bohm effect is *not* just a quantum mechanical effect, but also applies to classical electrodynamics. This is an example of the “correspondence principle” between the quantum domain and the macroscopic world.
- (4) A_z and E_z may be individually calculated from amplitude/phase data collected with a VNA. The algorithm has been noted in Eqs. (3a) and (3b).

- (5) Over far-field propagation tests, E_z drops off as $1/R^2$, while A_z diminishes as $1/R$. These are in agreement with anticipated results.
- (6) Communication of information should be possible by modulating $\mathbf{A}$ without transmitting TEM RF fields or power. This form of communication, using longitudinal $\mathbf{A}$, would be essentially covert in nature as it cannot be detected by a wire antenna.
- (7) Electromagnetic potentials are just as “real” (physically observable) as electromagnetic fields.
- (8) The $1/R$ dependence of A_z supports the gauge definition in Eq. (9). If charge/current continuity is asserted in the transmitting monopole, then the Lorenz gauge must be taken as the only gauge supporting physical propagation of longitudinal $\mathbf{A}$ waves (and also the scalar potential ϕ).
- (9) Detection of vector potential depends critically on plasma high-velocity electron response to $\mathbf{A}$. Valence electrons in metals are far too slow to respond to $\mathbf{A}$. This has always been observed historically, leading Jackson¹⁶ and others to conclude that vector potential is unobservable. This view must be questioned in view of the present work. Indeed, the concept of using an arbitrary gauge for mathematical convenience must now be re-evaluated, as there is no physical evidence supporting this.
- (10) By using the Lorenz gauge, $\mathbf{E}$ and $\mathbf{H}$ infer $\mathbf{A}$ and ϕ , and reciprocally.¹⁷ The Lorenz gauge provides proper Lorentz transformations in special relativity, which is a requirement if the potentials are observable.

Acknowledgment

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Appendix A. Theoretical Basis for Momentum Coupling in the Lorenz Gauge

The conjugate momentum density for the electron plasma in the tube may be written,

$$\mathbf{P}_{\text{conj}} = \rho_m \mathbf{v} + \rho \mathbf{A}, \quad (\text{A.1})$$

where $\rho_m = (m/e)\rho$ is the electron mass density and ρ is the charge density. The conjugate momentum relates to the interaction energy such that

$$\frac{d}{dt} \mathbf{P}_{\text{conj}} = -\nabla[\rho\phi - \mathbf{J} \cdot \mathbf{A}], \quad (\text{A.2})$$

where $\mathbf{J}$ is the current density and ϕ is the electric scalar potential. For this discussion, $\mathbf{J}$ and $\mathbf{A}$ have only $\hat{z}$ components and we write,

$$\frac{d}{dt} P_{\text{conj}z} = -\frac{\partial}{\partial z}[\rho\phi - J_z A_z]. \quad (\text{A.3})$$

Here we will demonstrate that $P_{\text{conj}z}$ is conserved (constant in time) by showing that the right-hand side of (A.3) is zero.

Charge continuity in the plasma requires that

$$\frac{\partial J_z}{\partial z} = -\frac{\partial \rho}{\partial t}. \quad (\text{A.4})$$

We assume that in addition to their constant DC values, J_z and ρ have a harmonic space-time dependence with frequency ω and wavenumber k_z . In the frequency domain, Eq. (A.4) may be written

$$(-jk_z)\tilde{J}_z = -(j\omega)\tilde{\rho}, \quad (\text{A.5})$$

$$\tilde{J}_z = c\tilde{\rho}. \quad (\text{A.6})$$

That is, the RF component of the current density is simply related to the RF component of the charge density.

In the Lorenz gauge, we have

$$\nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial \phi}{\partial t} = -\frac{1}{c^2} \frac{\partial \phi}{\partial t}. \quad (\text{A.7})$$

Again, for the $\hat{z}$ components only,

$$\frac{\partial A_z}{\partial z} = -\frac{1}{c^2} \frac{\partial \phi}{\partial t}. \quad (\text{A.8})$$

Changing once again to the frequency domain,

$$(-jk_z)\tilde{A}_z = -\frac{1}{c^2}(j\omega)\tilde{\phi}, \quad (\text{A.9})$$

$$\left(\frac{\omega}{c}\right)\tilde{A}_z = \left(\frac{\omega}{c^2}\right)\tilde{\phi}, \quad (\text{A.10})$$

$$\tilde{\phi} = c\tilde{A}_z. \quad (\text{A.11})$$

Then Eq. (A.3) may then be written as

$$\frac{d}{dt} P_{\text{conj}} = -\frac{\partial}{\partial z}[\rho\phi - J_z A_z] = -\frac{\partial}{\partial z} \left[\left(\frac{J_z}{c}\right) c A_z - J_z A_z \right] = 0. \quad (\text{A.12})$$

This leads immediately to

$$\frac{d}{dt}(\rho_m v_z + \rho A_z) = 0, \quad (\text{A.13})$$

$$\rho_m v_z + \rho A_z = \rho \left[\left(\frac{m}{e} \right) v_z + A_z \right] = \text{constant}. \quad (\text{A.14})$$

The bracketed term is zero (constant) if we choose

$$m\tilde{v}_z = -e\tilde{A}_z, \quad (\text{A.15a})$$

$$m\bar{v}_z = -e\bar{A}_z. \quad (\text{A.15b})$$

Equation (A.15a) describes a solution in ω for the RF components. Equation (A.15b) is a DC constraint for the mean electron velocity in the plasma tube as a function of the mean vector potential, which is due to the 200 mA of DC current flowing through the tube.

Appendix B. Interaction Energy Requirement for Full Momentum Coupling

To couple a momentum $\pm p$ to a plasma free electron,

$$p = eA, \quad (\text{B.1})$$

requires an available energy of

$$E = \frac{p^2}{2m_e} = \frac{(eA)^2}{2m_e}. \quad (\text{B.2})$$

This energy must be available from the interaction energy. Therefore,

$$\text{Interaction energy} = e(\mathbf{v}_{\text{drift}} \cdot \mathbf{A}) > \frac{(eA)^2}{2m_e}. \quad (\text{B.3})$$

Accordingly, to fully couple linear momentum for A_z requires a minimum electron drift velocity of

$$v_{\text{drift}} > \frac{eA}{2m_e}. \quad (\text{B.4})$$

For $A = 10^{-10}$ Wb/m, the plasma drift velocity must exceed 9 m/s. This requirement is abundantly satisfied by the low-temperature plasma used in this experiment, which has a modeled drift velocity¹² of about 40 km/s (Mach 120).

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